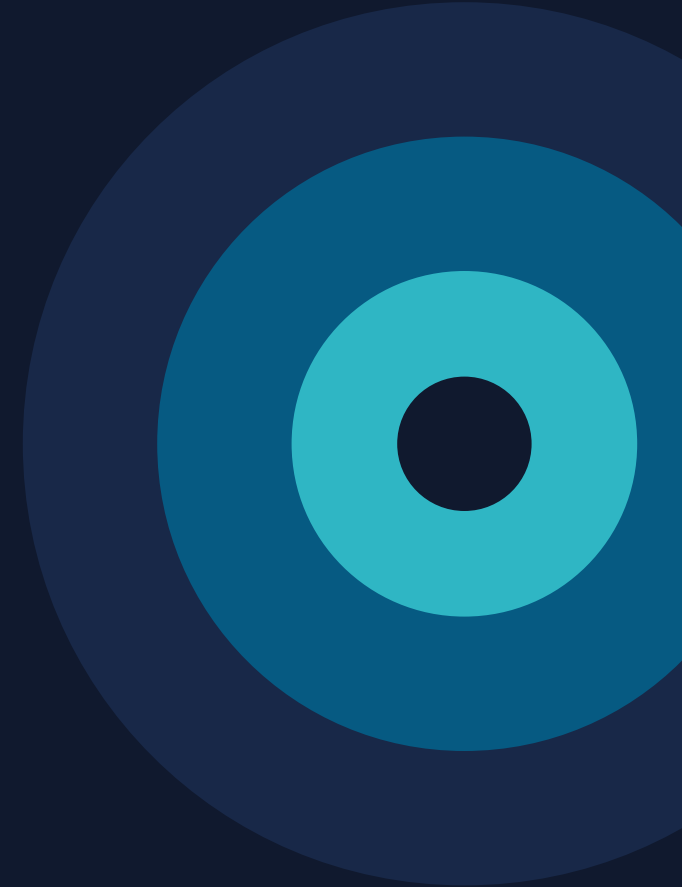


UNIT · ARTIFICIAL INTELLIGENCE

Intelligent Agents and Computer Vision

How machines perceive, decide, act — and see

First-Year B.Tech | Foundations of AI



Learning Objectives

- 1 Understand what an intelligent agent is, and why it matters in AI.
- 2 See how an agent perceives and acts within an environment.
- 3 Learn the basic structure and working of intelligent agents.
- 4 Differentiate between the five major types of intelligent agents.
- 5 Grasp the basic idea of computer vision and how machines interpret images.

What is an Intelligent Agent?

Agent

Anything that perceives its environment through sensors and acts upon it through actuators.

Intelligent Agent

An agent that acts rationally — it chooses the action expected to best achieve its goals, given what it has perceived.

Everyday analogy

Think of a security guard watching CCTV screens (sensors), deciding what's suspicious (reasoning), and pressing an alarm or opening a gate (actuators). That loop — watch, think, act — is exactly what an intelligent agent does.

Real-world examples

1

Self-driving car

Perceives road via cameras/LiDAR, decides steering & speed, acts on wheel & brakes

2

Chatbot

Perceives typed text, decides best reply, acts by generating a response

3

Robot vacuum cleaner

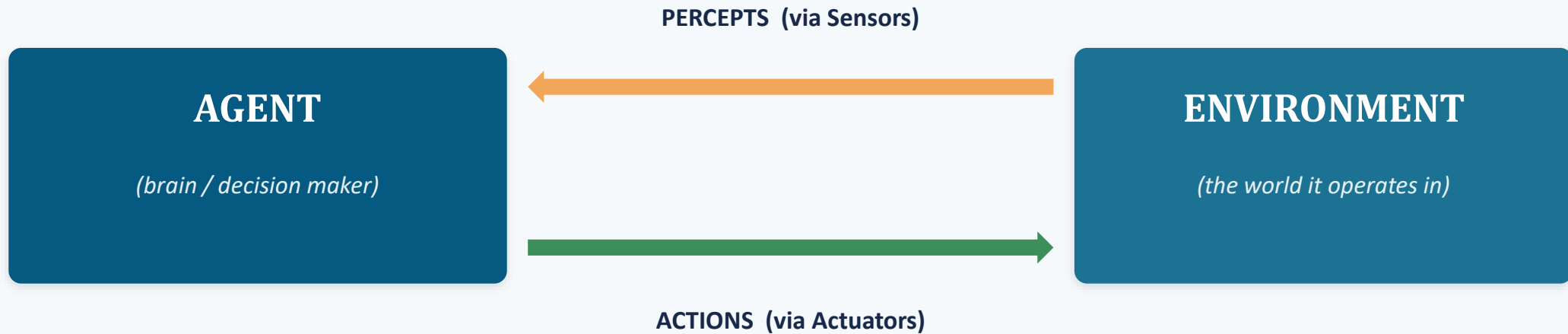
Perceives obstacles via bump/IR sensors, decides path, acts by moving & cleaning

4

Recommendation system

Perceives your clicks/history, decides relevant items, acts by showing suggestions

Agent–Environment Interaction



Percept

What the agent senses at one instant (e.g. a single camera frame)

Percept Sequence

The complete history of everything perceived so far

Action

What the agent does in response — a movement, output or decision

Sensors and Actuators



Sensors — how the agent perceives

Devices that let an agent capture information from the environment.

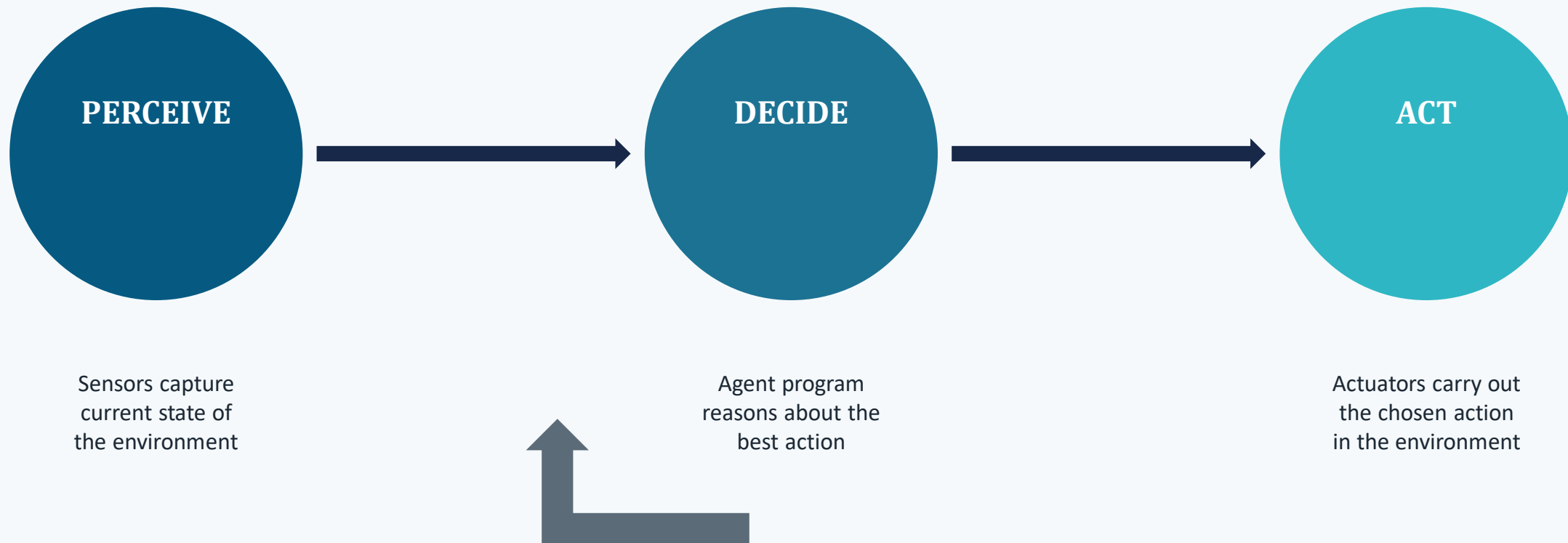


Actuators — how the agent acts

Devices/motors that let an agent carry out actions in the environment.

Agent	Sensors (input)	Actuators (output)
Human	Eyes, ears, skin, nose, tongue	Hands, legs, mouth (muscles)
Self-driving car	Camera, LiDAR, GPS, speedometer	Steering motor, brakes, accelerator
Robot vacuum	Bump sensor, IR sensor, camera	Wheels, suction motor, brush
Chatbot	Keyboard/mic (text or speech input)	Screen/speaker (text or speech output)

The Percept → Decide → Act Cycle



The cycle repeats continuously — every action changes the environment, producing new percepts

Agent Function vs Agent Program

Agent Function

An abstract, mathematical mapping from every possible percept sequence to an action.

f : Percepts* → Actions

Think of it as a giant lookup table — an idea, not something we actually build. It just defines correct behaviour.

Agent Program

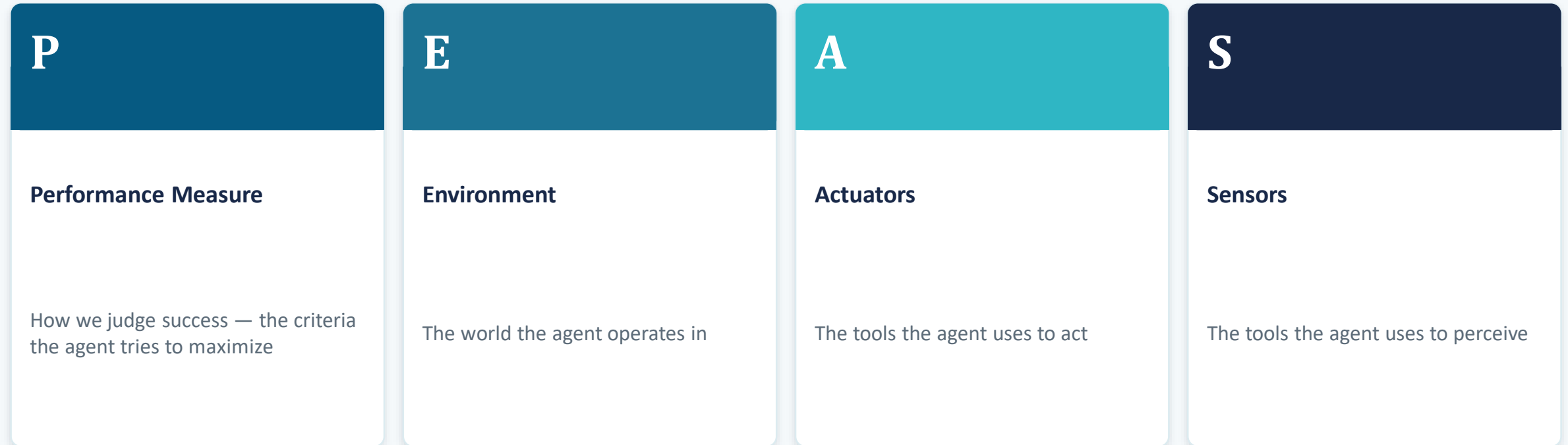
The real, concrete piece of software that runs on the agent's hardware and actually implements the agent function.

```
if (obstacle_ahead)
    turn();
else
    move_forward();
```

This is what a programmer actually writes and what runs inside the robot, car or chatbot.

PEAS Framework

Before building an agent, we describe its task environment using PEAS:



PEAS in Action — Worked Examples

	Self-Driving Car	Automated Vacuum Cleaner
Performance	Safety, speed, legality, comfort, fuel efficiency	Cleanliness, efficiency, battery life, coverage
Environment	Roads, traffic, pedestrians, signals, weather	Rooms, furniture, carpets, stairs, pets
Actuators	Steering, accelerator, brakes, horn, signals	Wheels/motors, brushes, suction, dustbin release
Sensors	Cameras, LiDAR, radar, GPS, speedometer	Bump sensors, cliff sensors, camera, dirt sensor

Structure of an Intelligent Agent



Architecture vs Program, and Decision-Making

	Agent Architecture	Agent Program
What it is	The physical platform: computer, sensors, actuators, memory	The software/logic that runs on that platform
Nature	Hardware (fixed)	Code (can be changed/updated)
Analogy	The human body (eyes, hands, brain hardware)	The mind/thoughts running inside that body
Relation	Agent = Architecture + Program	Same architecture can run different programs

Worked example: how a robot vacuum decides

Percept: bump sensor triggers → Agent program rule: "if bumped, turn 90° and move forward" → Action: motors turn wheels → New percept is generated, and the cycle continues.

Five Types of Intelligent Agents

As agents get more sophisticated, they add new capabilities on top of the previous type:



Each new type adds an ability the previous type lacked — think of it as five levels of increasing intelligence.

Simple Reflex Agent

Definition

Chooses actions using only the current percept, ignoring percept history entirely. It follows fixed condition–action rules: “if condition, then action.”

💡 **What's new here:** The very first, most basic decision rule: condition → action.

Real-world example

A thermostat: if room temperature < set point, turn heater ON; else turn it OFF.

Advantages

- Very simple to design and build
- Extremely fast to respond

Limitations

- No memory of the past
- Fails in partially observable environments

How it decides

Current Percept



`if (condition) → action`



Action Taken

Model-Based Reflex Agent

Definition

Keeps an internal state / “model” of the world, built from percept history, so it can handle situations that are only partially observable.

💡 **What's new here:** Adds **MEMORY** — an internal model of how the world works and what has been seen so far.

Real-world example

A self-driving car remembering the position of a car in its blind spot even after it can no longer see it directly.

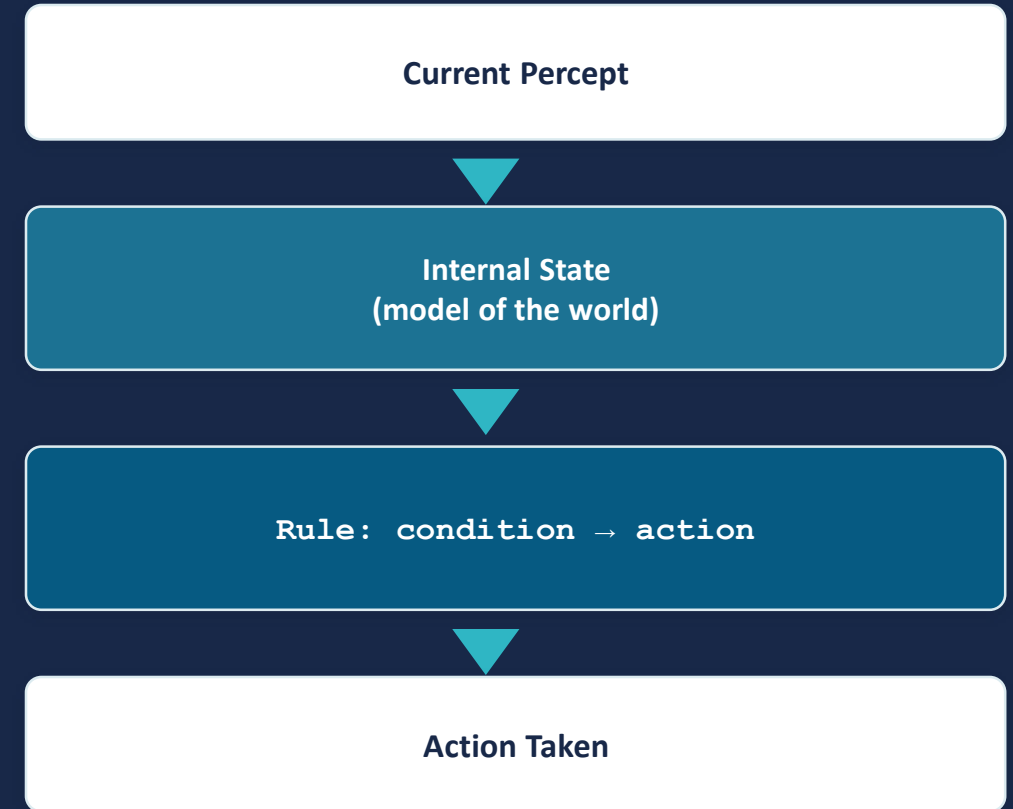
Advantages

- Handles partial observability
- Better decisions using history

Limitations

- Model can become outdated/wrong
- Still no explicit goals

How it decides



Goal-Based Agent

Definition

Acts to achieve a specific goal, not just to react. It considers the future — “what will happen if I do this?” — and picks actions that move it toward the goal.

💡 **What's new here:** Adds **GOALS** — a target/destination the agent is actively trying to reach, and simple planning to get there.

Real-world example

A GPS navigation app choosing a route: the goal is “reach the destination,” and it plans the best path.

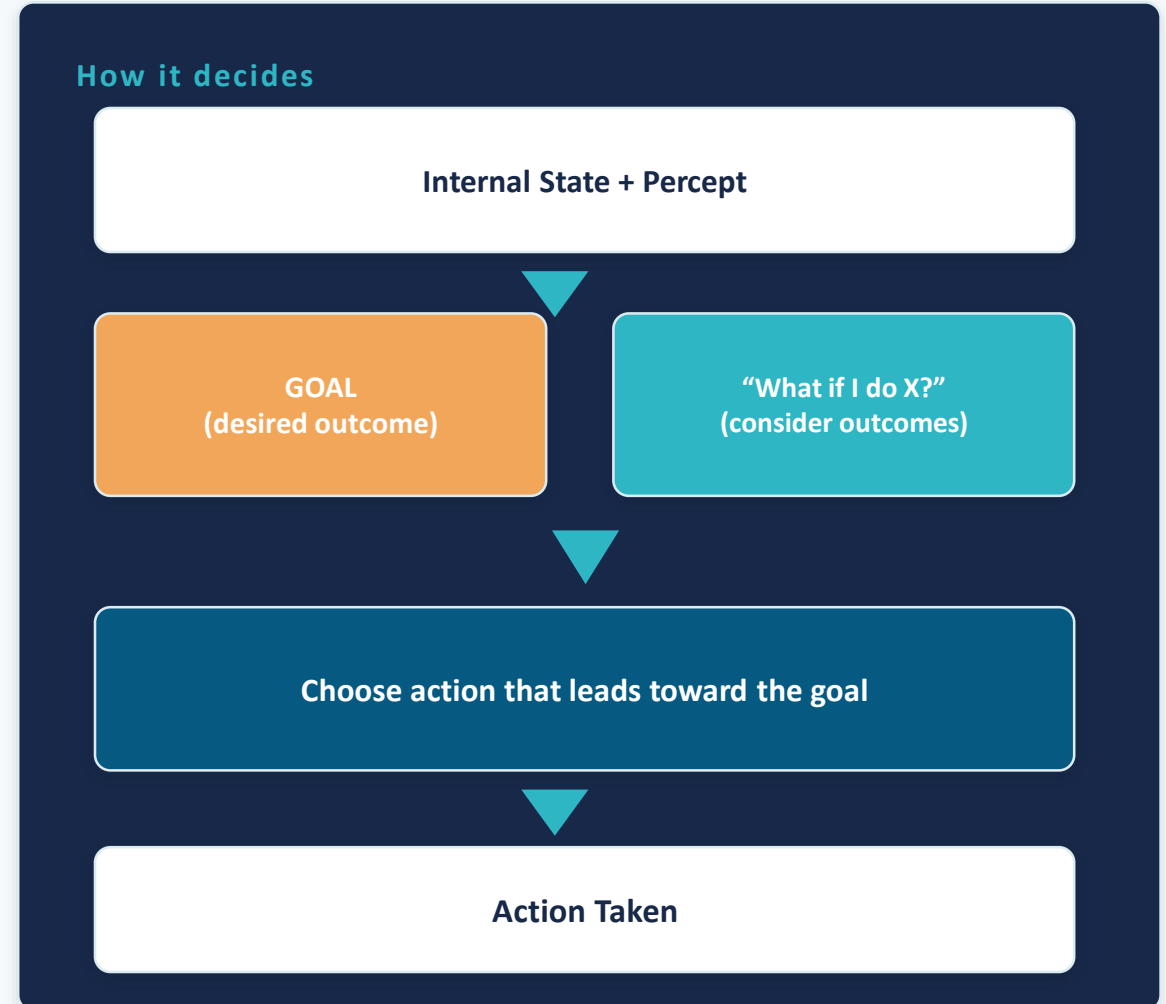
Advantages

- Flexible — goal can change
- Supports planning ahead

Limitations

- Doesn't rank multiple ways to reach the goal
- Can be slower / more computation needed

How it decides



Utility-Based Agent

Definition

Goes beyond simply reaching a goal — it measures HOW GOOD each possible outcome is using a utility function, and picks the action with the highest expected utility.

💡 **What's new here:** Adds a **UTILITY FUNCTION** — a numeric score of “goodness”, allowing the agent to choose the **BEST** option among many that all satisfy the goal.

Real-world example

GPS navigation choosing the fastest route among several valid routes to the same destination, balancing time, tolls and traffic.

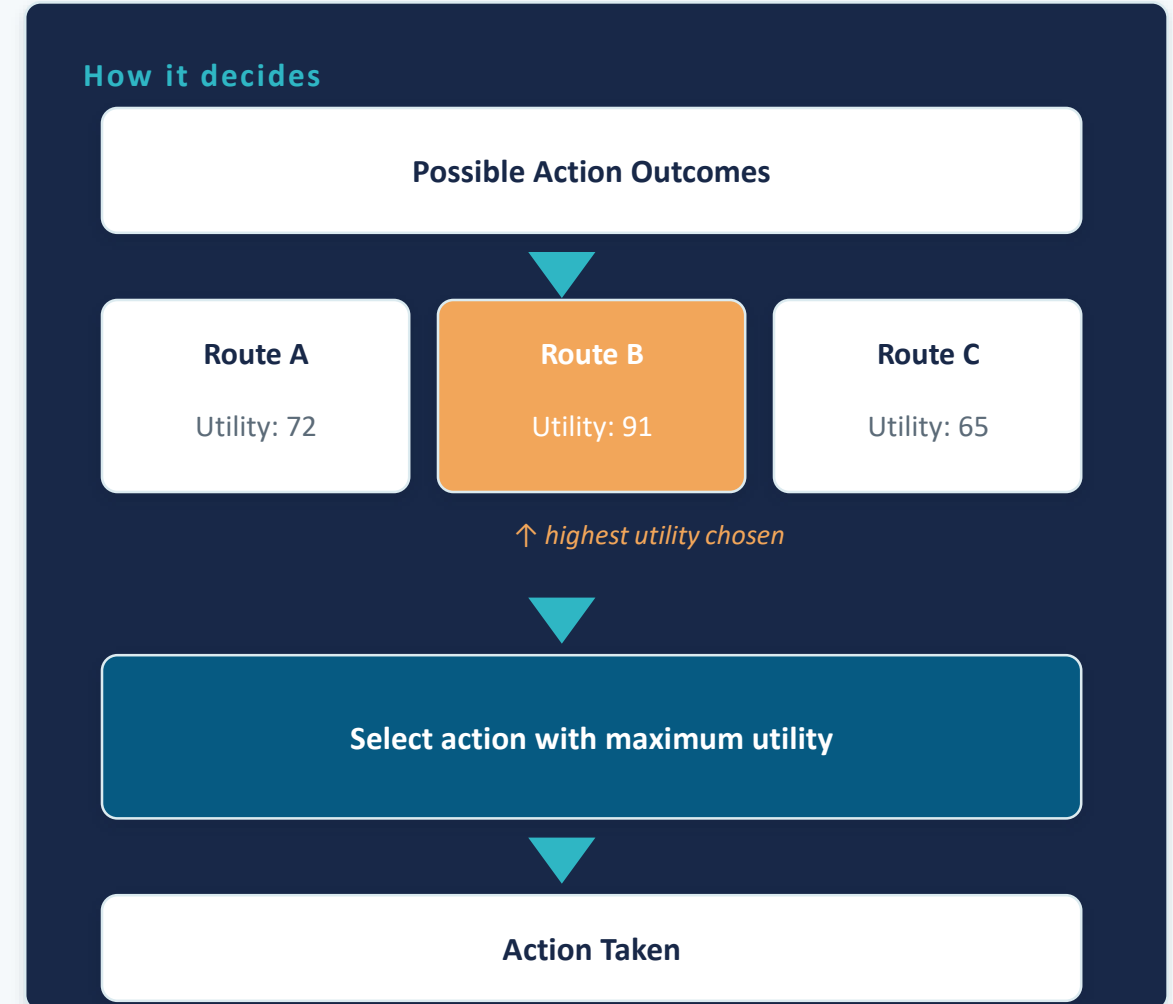
Advantages

- Compares multiple valid options
- Handles trade-offs (speed vs cost vs safety)

Limitations

- Designing a good utility function is hard
- More computation than simple goal-seeking

How it decides



Learning Agent

Definition

Can improve its own performance over time by learning from experience, without being explicitly reprogrammed for every new situation.

💡 **What's new here:** Adds **LEARNING** — a feedback loop that improves the agent program itself using a Critic, Learning Element and Problem Generator.

Real-world example

A recommendation system (Netflix/YouTube/Amazon) that gets better at suggesting content the more you interact with it.

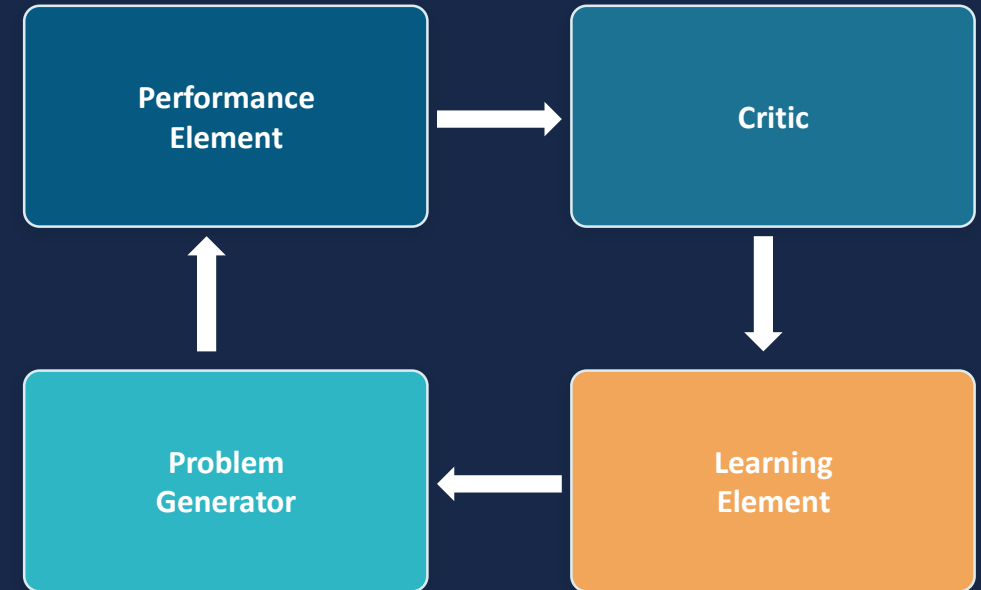
Advantages

- Adapts to new/unknown environments
- Improves automatically with more data

Limitations

- Needs data and time to learn
- Behaviour can be harder to predict/explain

How it decides



Feedback improves performance continuously

Critic tells the Learning Element how well the agent is doing; the Learning Element updates the Performance Element; the Problem Generator suggests exploratory actions to discover new knowledge.

Comparing the Five Agent Types

Agent Type	Uses Memory?	Has a Goal?	Compares Options?	Learns?	Example
Simple Reflex	No	No	No	No	Thermostat
Model-Based Reflex	Yes	No	No	No	Car sensing blind spot
Goal-Based	Yes	Yes	No	No	GPS reaching destination
Utility-Based	Yes	Yes	Yes	No	GPS picking fastest route
Learning	Yes	Yes	Yes	Yes	Netflix recommendations

SECTION 4

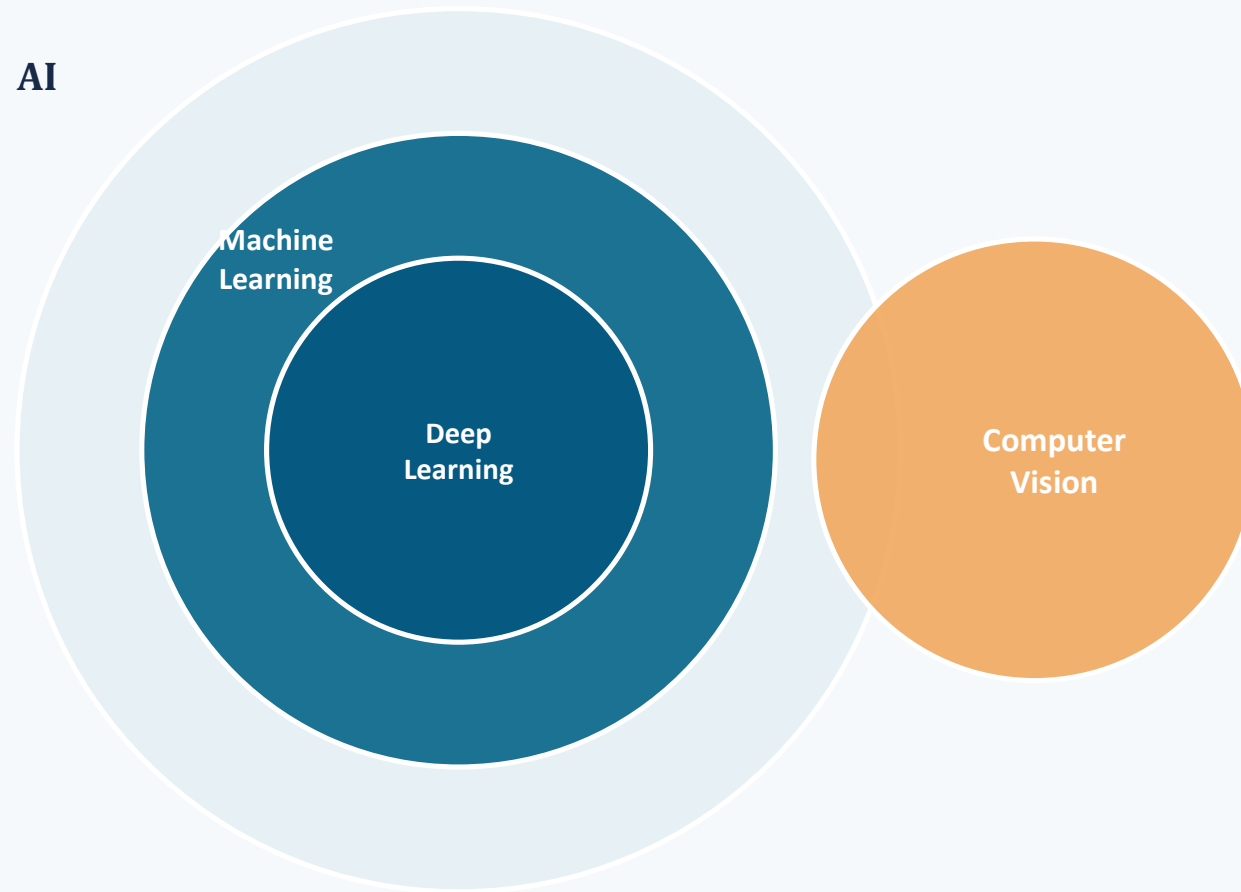
Introduction to Computer Vision

Teaching machines to “see” and make sense of images and video.

Computer Vision (CV) is a field of AI that enables computers to interpret and understand visual information from the world — just as our eyes and brain let us understand what we see.



How AI, ML, Deep Learning and CV Relate



AI (Artificial Intelligence)

The broad field: making machines act intelligently

ML (Machine Learning)

A subset of AI: learning patterns automatically from data

DL (Deep Learning)

A subset of ML using multi-layered neural networks

CV (Computer Vision)

An application area — often powered by DL — focused on visual data

How Humans See vs How Computers 'See'

Human Vision

- Eyes capture light and send signals to the brain
- Brain instantly recognizes objects, faces, context
- Understanding feels immediate and effortless
- Built on years of biological learning

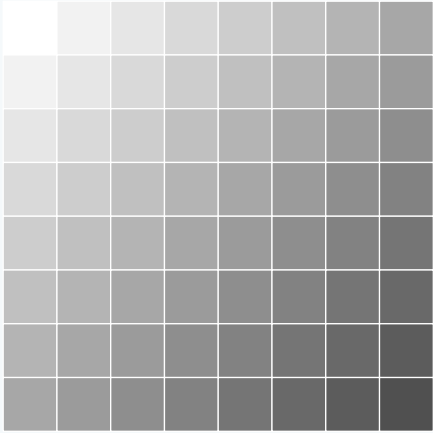
"That's a cat" — instantly, using intuition built over a lifetime.

Computer 'Vision'

- Camera captures an image as a grid of numbers
- No inbuilt understanding — just numeric data
- Algorithms/models process these numbers step-by-step
- "Learns" to recognize patterns from many examples

```
[ [23,45,12], [88,90,76], ... ] → after  
processing → "cat: 96% confidence"
```

How Computers Represent Images



Each tiny square is one pixel

Pixel

The smallest unit of a digital image — a single point of color/intensity.

Image Resolution

Number of pixels in an image (e.g. 1920×1080).
More pixels = more detail.

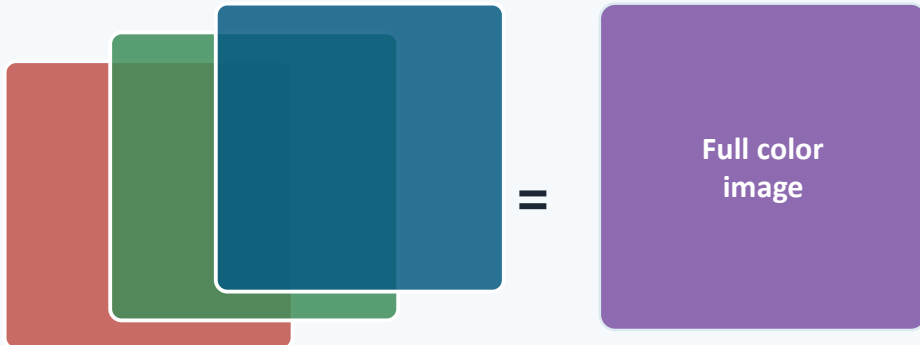
Grayscale Image

Each pixel has 1 value (0=black to 255=white) representing brightness only.

RGB Image

Each pixel has 3 values — Red, Green, Blue — combined to make any color.

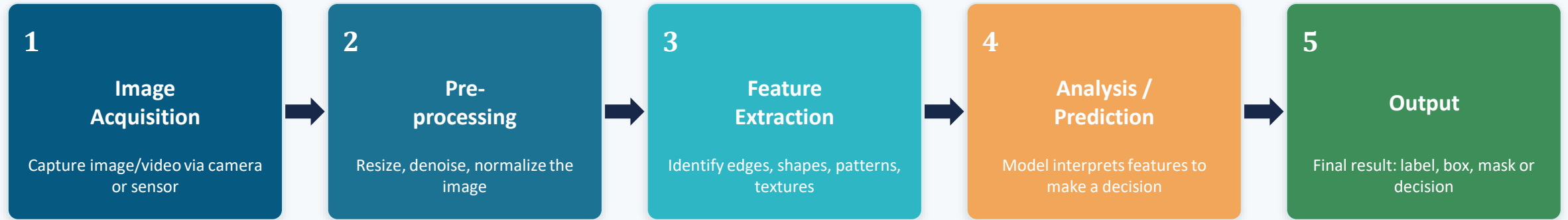
RGB channels stack to form one color image:



Key idea: Image as Numerical Data

A grayscale image is really a 2D matrix of numbers (rows $\times$ columns); an RGB image is 3 such matrices stacked together (height $\times$ width $\times$ 3). This numeric form is exactly what algorithms process.

The Basic Computer Vision Pipeline



Worked example: recognizing a handwritten digit

A scanned photo of “7” is captured (acquisition) → converted to grayscale and resized (pre-processing) → its strokes/edges are identified (feature extraction) → a model compares these features to known digit patterns (analysis) → the system outputs “7” (output).

Major Computer Vision Tasks

1

Image Classification

Assigns one label to the whole image

e.g. "This image is a dog"

2

Object Detection

Finds & locates multiple objects with bounding boxes

e.g. Boxes around every car in traffic

3

Image Segmentation

Labels every pixel — outlines exact object shape

e.g. Separating a tumor from tissue in a scan

4

Face Recognition

Identifies or verifies a specific person's identity

e.g. Unlocking your phone with your face

5

OCR (Text Recognition)

Reads and extracts text from images

e.g. Scanning a printed page into editable text

Real-World Applications of Computer Vision

1

Autonomous Vehicles

Detecting lanes, pedestrians, signs and other vehicles

2

Medical Imaging

Spotting tumors, fractures, and anomalies in scans

3

Security & Surveillance

Face recognition, intrusion & anomaly detection

4

Agriculture

Crop health monitoring, weed & pest detection via drones

5

Smartphones

Face unlock, portrait mode, document scanning, AR filters

6

Manufacturing

Automated defect detection on production lines

The Role of CNNs and Deep Learning in CV

Why not just hand-code the rules?

Early computer vision relied on humans manually designing rules to detect edges, corners and shapes. This worked poorly for messy, real-world images (different lighting, angles, backgrounds).

Deep Learning changed this: instead of hand-coding features, a Convolutional Neural Network (CNN) learns the best features automatically from thousands of example images.

Analogy: instead of telling a child exact rules for recognizing a cat, you show it thousands of cat pictures until it learns the pattern itself.

How a CNN loosely works

- 1 Scans the image in small patches, looking for simple patterns (edges, colors)
- 2 Combines simple patterns into more complex ones (shapes, textures)
- 3 Combines those into even higher-level concepts (eyes, wheels, leaves)
- 4 Final layer makes a prediction (e.g. “cat”, “car”, “leaf disease”)

No advanced math needed here — just remember: CNNs learn layers of increasingly complex visual patterns automatically from data.

KEY TAKEAWAYS

What to Remember from This Unit

- ✓ An intelligent agent perceives via sensors and acts via actuators, following the Percept → Decide → Act cycle.
- ✓ PEAS (Performance, Environment, Actuators, Sensors) describes any task environment precisely.
- ✓ Agent = Architecture (hardware) + Program (software) — the two are different things.
- ✓ Five agent types, in order of sophistication: Simple Reflex → Model-Based → Goal-Based → Utility-Based → Learning.
- ✓ Computer Vision lets machines interpret visual data — represented as pixels, often as RGB numeric matrices.
- ✓ The CV pipeline: Acquisition → Preprocessing → Feature Extraction → Analysis → Output.
- ✓ Modern CV relies heavily on CNNs / Deep Learning to learn visual features automatically from data.

Short Quiz — Review Questions (1–5)

1 Define an intelligent agent in your own words. What role do sensors and actuators play?

2 What does PEAS stand for? Give the PEAS description for a hospital patient-monitoring agent.

3 Explain the Percept → Decide → Act cycle with a simple example of your own.

4 What is the difference between an agent's architecture and its agent program?

5 Name the five types of intelligent agents in order of increasing sophistication.

Short Quiz — Review Questions (6–10)

6

A robot vacuum remembers where it has already cleaned. Which agent type is this, and why?

7

How does a utility-based agent differ from a goal-based agent? Give an example.

8

Define computer vision. How is it related to AI and Machine Learning?

9

What is a pixel? What is the key difference between a grayscale image and an RGB image?

10

List the five stages of the basic computer vision pipeline, and name one real-world CV application for each of: classification, detection, segmentation, face recognition, OCR.